Phase D6 and Series D Final Closure

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Attribution

> *This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).*

Abstract

This note is D6.15 — the fifteenth and final note in Phase D6, and the final note in Series D. Phase D6 produced twelve boundary cases organized into three types: Type A (minimum conditions), Type B (maximum conditions), and Type C (definiteness conditions), concluded by a consolidated boundary map in D6.14. Series D as a whole comprises 206 notes across six phases (#460–#665), making it the largest single-paper derivation series in the CKS corpus. The six phases together establish that Paper 3's inter-Self coordination architecture is operationally specified, failure-mode-characterized, composition-interaction-documented, compliance-testable, and scope-bounded. This closure note summarizes what Phase D6 contributed, states what the complete Series D corpus accomplished across its six phases, and describes the transition to Series CC, which supplies the cross-paper inheritance layer that Series D's internal focus left as a remaining gap.

1. Phase D6 in context

Phase D6 is the sixth and final phase of Series D. Its purpose is scope boundary precision: establishing, for Paper 3's inter-Self coordination architecture, the exact conditions at which a given scenario falls within the architecture's scope and the conditions at which it falls outside. Scope ambiguity is an adversarial vulnerability — a commitment whose application range is imprecise can be narrowed by a challenger who argues that the target scenario lies just outside it. Phase D6 forecloses that strategy by making the boundaries explicit.

Phase D6 contains fifteen notes. Note D6.01 established the three-type taxonomy that organizes the twelve cases. Notes D6.02 through D6.13 produced the twelve boundary cases themselves.

Note D6.14 presented a consolidated boundary map summarizing all twelve cases as a reference for downstream work. This note, D6.15, closes Phase D6 and Series D.

2. The twelve boundary cases: three types

Type A — Minimum conditions (three cases). Type A cases establish the floor: the least that an event, substrate, or participant must satisfy for the architecture's commitments to apply. D6.02 addressed the minimum valid FAI event — the conditions below which an inter-Self exchange does not constitute a Full Aspect Integration event under the architecture's commitments. D6.03 addressed the minimum valid shared substrate — the structural conditions a substrate must meet before it qualifies as the architectural object of inter-Self coordination. D6.04 addressed the minimum valid participant — what a Self must satisfy to count as a participating party in an FAI event rather than a passive or downstream consumer. Type A cases address the adversarial strategy of arguing that no real instance satisfies the architecture; by specifying the minimum precisely, they establish that the architecture has a reachable floor.

Type B — Maximum conditions (three cases). Type B cases establish the ceiling: the most that sharing scope, cardinality, and governance depth can reach while remaining within the architecture's commitments. D6.05 addressed maximum sharing scope — the broadest substrate content that can be made jointly accessible under a single FAI event without violating the home-perimeter governance each participating Self retains. D6.06 addressed maximum cardinality — the upper bound on the number of Selves that can participate in a single FAI event under the architecture's joint-authority and conflict-handling mechanisms. D6.07 addressed maximum governance depth — the deepest recursive application of governance commitments that remains internally consistent under Paper 3's Claim 5 recursive-applicability commitment. Type B cases address the adversarial strategy of arguing that the architecture claims infinite scope and is therefore unfalsifiable; they establish that the scope has a reachable ceiling.

Type C — Definiteness conditions (six cases). Type C cases establish partition lines between the architecture's scope and adjacent concepts that share surface features but differ in architectural commitment. D6.08 addressed the FAI/foil boundary — what distinguishes an FAI event from coordination patterns that exchange content without satisfying the architecture's full-merge default and substrate-mediation commitments. D6.09 addressed the intra/inter-Self boundary — what distinguishes coordination within a single Self's perimeter from coordination that crosses it and therefore engages Paper 3's commitments rather than Paper 2's. D6.10 addressed the temporary/persistent boundary — what distinguishes a transient exchange from a persistent FAI event that generates the substrate state and evolution-feed consequences the architecture specifies. D6.11 addressed the cooperation/competition boundary — what distinguishes Selves whose coordination is cooperative in the sense that generates full FAI commitments from Selves whose interaction is adversarial. D6.12 addressed the FAI/merger-and-acquisition boundary — what distinguishes a Full Aspect Integration event, which preserves both Selves' perimeters, from a Self-level merger that collapses them. D6.13 addressed the bilateral/population-scope boundary — what distinguishes a bilateral FAI event from a population-scale coordination event under Claim 6's scope. Type C cases address the adversarial strategy of collapsing the architecture into an adjacent concept; by specifying six partition lines, they establish that the architecture is positively differentiated from its closest neighbors in each dimension.

3. The consolidated boundary map (D6.14)

Note D6.14 served as Phase D6's reference artifact. It presented all twelve boundary cases on a single consolidated map organized by type, with each case's scope-inclusion and scope-exclusion conditions stated in parallel form. The map is not an additional boundary case; it is a summary that makes the twelve cases jointly navigable by downstream notes in Series CC, Series T, and any subsequent derivation work. Its function parallels the consolidated test library that Phase D5 produced: both are reference artifacts that enable later phases to draw on earlier work without re-establishing it case by case.

4. What the complete Series D corpus accomplished

Series D comprises 206 notes across six phases, spanning note numbers #460 through #665. It is the largest single-paper derivation series in the CKS corpus, reflecting Paper 3's architectural complexity: each of its six claims inherits from and extends prior-paper commitments at inter-Self scope, generating more derivation surface per claim than either prior paper produced.

Phase D0 (6 notes, #460–#465) established Paper 3's six claims as explicit prior-art anchors at the paper-claim level. Phase D0 is the parent layer for all subsequent phases: every sub-commitment, operational variant, anti-pattern, composition pair, test, and boundary case in Series D traces to one of the six D0 claim anchors.

Phase D1 (30 notes, #466–#495) decomposed each claim into its foundational sub-commitments — the subordinate architectural decisions that each claim commits to, and that a party would need to independently develop to practice the architecture without engaging the claim anchor directly.

Phase D2 (80 notes, #496–#575) stated the operational requirements for every Paper 3 commitment: the specific system behaviors, preconditions, postconditions, and invariants that a compliant implementation must satisfy. At 80 notes, Phase D2 is the largest single phase in Series D. It addresses the adversarial strategy of arguing that Paper 3's commitments are too abstract to have operational consequences; the 80 notes collectively demonstrate that each commitment is operationally specified.

Phase D3 (30 notes, #576–#605) produced a 27-anti-pattern taxonomy, with each anti-pattern receiving a standalone note that names the failure mode, specifies its observable symptoms, and states the minimal condition under which it occurs. Phase D3 addresses the adversarial strategy of arguing that the architecture’s failure modes are uncharacterized: a challenger who introduces a failure mode that D3 has already named and characterized cannot claim it as novel territory.

Phase D4 (30 notes, #606–#635) characterized 26 composition pairs — interactions between distinct Paper 3 commitments, and interactions between Paper 3 commitments and Paper 1 or Paper 2 commitments. Phase D4 addresses the adversarial strategy of arguing that multi-commitment combinations are novel: a challenger who combines two commitments that the architecture already holds cannot claim the combination as independent invention.

Phase D5 (15 notes, #636–#650) established a 60-test operational test library, with each note containing multiple binary-testable conditions derived from Paper 3’s commitments. Phase D5 addresses the adversarial strategy of arguing that the architecture’s commitments are untestable: each test is stated as a binary condition that a given system either satisfies or does not, making compliance assessment concrete and replicable.

Phase D6 (15 notes, #651–#665) established 12 boundary cases in the three-type taxonomy, plus the consolidated boundary map in D6.14. Phase D6 addresses the adversarial strategy of arguing that the architecture’s scope is either too narrow to be relevant or too broad to be specific; the 12 cases establish that the scope is precisely defined at both floor and ceiling and that the architecture is positively differentiated from its closest neighbors.

Together the six phases establish that Paper 3’s inter-Self coordination architecture is operationally specified (D2), failure-mode-characterized (D3), composition-interaction-documented (D4), compliance-testable (D5), and scope-bounded (D6), resting on a foundation of named claim anchors (D0) and foundational sub-commitment decompositions (D1). Series D does not establish that Paper 3’s architecture is the only or best approach to inter-Self coordination; it establishes that the architecture’s commitments are publicly documented operational prior art that any party practicing the architecture would encounter.

5. What Series D does not cover: the transition to Series CC

Series D is organized around Paper 3 internally. Its 206 notes derive Paper 3’s commitments, characterize their operational requirements and failure modes, document their composition interactions, establish their testability, and bound their scope. Series D does not formally trace the inheritance edges between Paper 3’s commitments and the commitments established in Papers 1 and 2. That tracing is the work of Series CC.

The inheritance gap was identified during Series D's development. Paper 3's claims do not arise independently: each inherits specific prior commitments from Paper 1 (the human-governed substrate architecture) and Paper 2 (the instinct/reasoning separation and intra-Self evolution model), and extends those inherited commitments to the inter-Self scope. The inheritance is acknowledged within Paper 3's own per-claim accounting, but it has not been formalized as standalone prior-art notes. Series CC supplies that formalization.

Series CC will comprise approximately 40 notes across two subseries. Phase CC.1 (approximately 20 notes, #666–#685) produces cross-paper inheritance notes tracing each Paper 3 commitment to its Paper 1 antecedent — substrate identity at inter-Self scope inheriting from Paper 1's substrate commitment, human authority over the shared substrate inheriting from Paper 1's human-governed commitment, and so on across the inheritance edges the architecture identifies. Phase CC.2 (approximately 20 notes, #686–#705) produces cross-paper inheritance notes tracing each Paper 3 commitment to its Paper 2 antecedent — aspect-as-exchange-unit inheriting from Paper 2's intra-Self aspect model, FAI pattern variants inheriting from Paper 2's intra-Self combination primitive, and the remaining Paper 2 inheritance edges through to the evolution-feed and governance commitments.

The strategic function of Series CC is different from Series D's. Series D establishes that Paper 3's commitments are operationally specific and publicly documented. Series CC establishes that those commitments are not free-standing inventions: they inherit from a prior-art chain extending through Papers 1 and 2. A challenger who argues that Paper 3's architecture is novel in a way that bypasses Series D's prior art must also engage Series CC's explicit tracing of what Paper 3 inherits — limiting the territory in which novelty claims can be constructed.

6. Conclusion

D6.15 closes Phase D6, which produced twelve boundary cases in three types and a consolidated boundary map, completing the scope-boundary precision layer for Paper 3's inter-Self coordination architecture. Phase D6's closure marks the completion of Series D: 206 notes across six phases (#460–#665), providing operational requirements (Phase D2), failure-mode characterization (Phase D3), composition-interaction documentation (Phase D4), compliance-testable verification (Phase D5), and scope-boundary precision (Phase D6), resting on the claim anchors and foundational sub-commitments of Phases D0 and D1.

Series D is the most operationally comprehensive single-paper derivation series in the CKS corpus. Its scope reflects Paper 3's architectural complexity, and its purpose is specific: to establish, as publicly documented prior art, that Paper 3's inter-Self coordination architecture is operationally specified and scope-bounded in ways that address the most common adversarial challenges. What Series D does not do — formally trace the inheritance edges from Papers 1 and 2 — is what Series CC will do, beginning with note #666.

Source papers

Li, W. (2026a). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026b). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026c). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration.* April 2026. ORCID: 0009-0004-8065-3235.

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